MINI PROJECT LOGBOOK

Gap-Aware Retrieval (GAR)

A Gap-Aware Retrieval Algorithm for Iterative Retrieval-Augmented Generation

Project domain: Information Retrieval · Retrieval-Augmented Generation · NLP · Iterative Evidence Retrieval

Document type: Mini-project technical logbook / research-development record

Status: Proposed research formulation; empirical validation and prior-art verification are still required

Prepared as a structured record of the project journey, from the initial RAG/IR problem through literature exploration, formulation of GAR, mathematical modeling, experimental design, and the planned validation protocol.

Important note for submission: This logbook is based on the uploaded project draft and its cited literature. It deliberately describes GAR as a proposed formulation rather than claiming that the formulation is already proven novel or superior. Replace institutional fields, student names, guide name, dates, and actual experimental results before submission.

1. Project Identity and Starting Point

The uploaded project draft defines Gap-Aware Retrieval (GAR) as a retrieval algorithm intended for iterative Retrieval-Augmented Generation (RAG). The central observation is that ordinary retrieval systems usually rank documents by relevance to the original question. In an iterative setting, however, some parts of the information need may already have been satisfied, while other information remains missing. The draft therefore asks whether explicitly modeling the current information gap can improve the next retrieval decision. [Project draft, pp. 1–3]

The project is positioned inside the MIGRES framework, which stands for Missing Information Guided Retrieve-Extraction-Solving. MIGRES identifies information that is still missing after evidence has been collected and uses that state to guide another retrieval step. GAR is not presented as a replacement for MIGRES; it is the proposed retrieval component inside that larger iterative loop. [Project draft, pp. 1–3]

The starting point is therefore not a completely new RAG architecture. It is a narrower research question: if an iterative system already knows (i) the original question, (ii) evidence collected so far, and (iii) what information is still missing, can the retrieval-ranking function use all three signals rather than only query–document relevance?

1.1 The problem in one sentence

Given a question, previously collected evidence, and a current set of missing-information requirements, rank candidate documents so that the next retrieved documents are relevant, useful for the remaining gap, and non-redundant with what is already known.

1.2 Why this is a research problem

- One-shot retrieval and iterative retrieval have different states: after each iteration, the system has more evidence than it had before.
- A document can be highly relevant to the original question but contribute little new information at the current iteration.
- A useful retrieval system should therefore account for the information that remains unresolved, not only the semantic similarity between the original question and a candidate document.
- Any proposed method must be compared against strong lexical, dense, late-interaction, and neural-sparse retrieval baselines under the same experimental conditions.

2. The Project Story: How the Idea Emerged

2.1 Stage 1 — Starting from Retrieval-Augmented Generation

The initial intellectual setting was RAG. RAG systems connect a language model with an external document collection so that information can be retrieved at inference time. This makes retrieval a critical component: the generator can only use external evidence that the retrieval stage makes available.

The first question was therefore conventional: how can a system retrieve the most relevant documents for a question? This naturally leads to established information-retrieval methods such as BM25, dense passage retrieval, late-interaction retrieval, and neural sparse retrieval.

2.2 Stage 2 — Realizing that the project is iterative

The next conceptual step was to examine multi-hop and iterative information seeking. In a multi-hop problem, one document may provide one fact while another document provides a second fact needed to connect the reasoning chain. The retrieval problem is no longer simply 'find documents about the question'; it becomes 'find the next evidence needed to complete the answer.'

This is where the MIGRES paper becomes the direct starting point for the present project. The uploaded draft identifies MIGRES as the surrounding framework and explicitly states that MIGRES finds what is missing, while GAR determines which documents are most useful for obtaining that missing information. [Project draft, pp. 2–3]

2.3 Stage 3 — The information-gap insight

Suppose a question requires two facts, A and B. If the first retrieval step already provides A, a conventional relevance-based retriever may continue returning documents about A because they remain highly similar to the original question. An iterative system instead has a new retrieval state: A is known and B is missing.

```
Question → retrieve A → identify missing B → retrieve B → answer
```

This simple example motivates the use of a gap variable G_t. The retrieval decision at iteration t is conditioned on the original question q, previously collected evidence E_t, and the missing-information state G_t.

2.4 Stage 4 — From intuition to a scoring function

The project then decomposes usefulness into three signals: relevance, gap coverage, and novelty. Relevance asks whether a document is semantically related to the question and current gap. Coverage asks whether the document contains content corresponding to the missing requirement. Novelty asks whether the document adds information that is not already represented in the collected evidence.

The proposed formulation combines the three normalized signals into a weighted score. The uploaded draft is explicit that the individual concepts are grounded in prior work; the research contribution is the proposed unified retrieval formulation and its controlled evaluation, not the claim that relevance, coverage, or novelty were individually invented here. [Project draft, pp. 1–2, 6]

2.5 Stage 5 — Turning the idea into an experiment

The experimental principle is to change the retrieval component while keeping the surrounding MIGRES pipeline as fixed as the original implementation permits. The draft therefore proposes comparisons with BM25, DPR, Contriever, ColBERTv2, and SPLADE, along with ablations that remove relevance, coverage, or novelty. [Project draft, pp. 3, 18–19]

The project consequently becomes a testable scientific question rather than only an equation: does state-aware ranking improve retrieval and downstream answer quality in iterative information-seeking tasks?

3. Problem Statement

Problem statement: Conventional retrieval methods primarily rank candidate documents according to their relevance to a query. In iterative RAG and multi-hop information-seeking tasks, the system has a changing evidence state: some information has already been collected while other information remains missing. A retriever that ignores this state can spend retrieval capacity on redundant or already-satisfied information. The project therefore investigates a retrieval algorithm that explicitly incorporates the current information gap and accumulated evidence into document ranking.

4. Objectives

1. Formulate a retrieval-ranking algorithm for iterative RAG using question relevance, information-gap coverage, and evidence novelty.
2. Integrate the proposed retriever with the MIGRES iterative retrieval framework without changing unrelated components wherever the original implementation permits.
3. Compare GAR against lexical, dense, late-interaction, and neural-sparse retrieval baselines.
4. Evaluate both retrieval quality and downstream question-answering quality.
5. Determine the contribution of each GAR component through controlled ablation studies.
6. Measure computational cost, candidate-pool sensitivity, retrieval latency, and the number of model calls.
7. Document the complete experimental configuration so that the results can be reproduced.

5. Research Questions

ID	Research question
RQ1	Does GAR improve retrieval quality over the MIGRES + BM25 baseline?
RQ2	How does GAR compare with dense, late-interaction, and neural-sparse retrieval methods?
RQ3	Does replacing the baseline retriever with GAR improve downstream MIGRES answer quality?
RQ4	What is the contribution of relevance, gap coverage, and novelty?
RQ5	Does the behavior generalize across different multi-hop and knowledge-intensive datasets?
RQ6	What computational cost is introduced by GAR?
RQ7	Can the retrieval improvement be obtained without adding unnecessary LLM calls?

6. Literature Trail

The literature trail should be understood as a chain rather than a flat bibliography. Classical IR provides the ranking foundations. Dense and late-interaction retrieval provide modern semantic retrieval mechanisms. Multi-hop and iterative retrieval establish the need for changing retrieval state. Query reformulation and active retrieval show that the retrieval input can change over time. Diversity and novelty methods establish the importance of avoiding redundant evidence. RAG evaluation work provides the methodology for testing the complete system.

6.1 Direct starting paper

Wang et al. (2025), “LLMs Know What They Need: Leveraging a Missing Information Guided Framework to Empower Retrieval-Augmented Generation.” The project draft explicitly identifies this work as the MIGRES framework on which GAR is evaluated. [Project draft, pp. 1–3; Ref. 1]

6.2 Iterative evidence retrieval

Yadav, Bethard, and Surdeanu (2020) provide an iterative evidence-retrieval perspective. Their work is especially relevant to the coverage idea because later retrieval can focus on information not adequately covered by existing evidence. [Project draft, p. 6; Ref. 2]

6.3 Relevant information gain and redundancy

Pickett et al. (2024) motivate retrieval that avoids wasting limited context on redundant information while preserving relevance. This line of work is directly related to the novelty term in GAR. [Project draft, p. 6; Ref. 3]

6.4 Classical and neural retrieval

BM25 provides the lexical baseline; DPR provides supervised dense retrieval; Contriever provides unsupervised dense retrieval; ColBERTv2 provides token-level late interaction; and SPLADE provides neural sparse retrieval. These methods cover substantially different retrieval representations and therefore provide a useful baseline family. [Project draft, pp. 4–6]

6.5 Additional literature that should be read before a paper submission

The extended reading set in this logbook includes RAG foundations (REALM, RAG, RETRO, Atlas), dense retrieval (ANCE, E5), query transformation (HyDE, Query2doc), iterative reasoning/retrieval (PullNet, HopRetriever, IRCoT), adaptive retrieval (FLARE, Self-RAG, CRAG, Adaptive-RAG), novelty/diversity (MMR), retrieval benchmarks (BEIR, KILT), and RAG evaluation (RAGAS, ARES, RAGTruth). These references are included to make the literature review broad enough to support a serious prior-art and experimental discussion.

7. Mathematical Foundations

7.1 Information retrieval as ranking

Let q denote a query and D = {d_1, d_2, …, d_n} denote the searchable document collection. A retrieval system maps the query and corpus to an ordered list of documents.

$$(q, D) \rightarrow d_1, d_2, \ldots, d_k$$

The central difference between retrieval algorithms is the scoring function used to determine which documents are placed near the top of the ranking.

7.2 Iterative retrieval state

At iteration t, the proposed system maintains three pieces of state. q is the original question, E_t is the evidence collected before iteration t, and G_t = {g₁, …, g_m} is the set of information requirements that remain unresolved.

$$State_t = (q, E_t, G_t, D)$$

The critical change is that retrieval is no longer modeled as only S(d | q). It becomes S(d | q, E_t, G_t). This means that the same candidate document can receive a different score at different iterations because the evidence and missing-information state have changed.

7.3 Text representation

GAR requires a representation function that maps text to a fixed-dimensional vector.

$$\varphi : T \rightarrow R^p$$

The project draft specifies the Sentence-Transformers model multi-qa-mpnet-base-cos-v1 as the fixed embedding model. It maps text to 768-dimensional vectors, and document vectors are computed once and stored in an approximate nearest-neighbor index. [Project draft, p. 8]

The important methodological point is that the embedding model is treated as a fixed representation component in the primary GAR experiment; GAR itself is not trained end-to-end in

the proposed setup.

7.4 Cosine similarity

For two text objects x and y, cosine similarity measures the angle between their embedding vectors.

$$\cos(x,y) = [\varphi(x)^T \varphi(y)] / [||\varphi(x)||_2 \, ||\varphi(y)||_2]$$

The draft converts this into a [0,1] score for convenience:

$$s(x,y) = \tfrac{1}{2} [1 + \cos(x,y)]$$

The conversion does not change the ranking induced by cosine similarity; it simply shifts and rescales the value to a non-negative normalized range.

8. GAR Mathematical Model

8.1 Gap weights

Suppose the current gap contains m requirements, g_1 through g_m. Each requirement receives a non-negative weight w_i. In the first implementation, all requirements have equal weight.

$$w_i \geq 0, \; \Sigma_i w_i > 0$$

$$w_i = 1 \text{ for all } i \text{ (initial implementation)}$$

The equal-weight setting avoids introducing another learned or manually tuned component. A later study could investigate learned or confidence-weighted gap importance, but that is outside the primary formulation.

8.2 Gap relevance

First, compute the weighted semantic similarity between a candidate document d and the current missing-information requirements.

$$s_G(d,G_t) = [\Sigma_i w_i \, s(d,g_i)] / [\Sigma_i w_i]$$

This gives one normalized score representing how semantically aligned the document is with the current information gap.

The system then combines relevance to the original question with relevance to the gap.

$$R(d,q,G_t) = \lambda \, s(d,q) + (1-\lambda) \, s_G(d,G_t), \; 0 \leq \lambda \leq 1$$

When λ is close to 1, the original question receives more weight. When λ is close to 0, the current gap receives more weight. This parameter therefore controls how strongly the retrieval process shifts from the global question toward the local missing-information state.

8.3 Gap coverage

Semantic similarity alone can be insufficient because a candidate may be generally related to a gap without actually covering the required terms or concepts. The draft therefore introduces a lexical coverage signal. Let τ(x) denote the set of normalized content tokens produced by the fixed preprocessing pipeline.

$$c(d,g_i) = |\tau(d) \cap \tau(g_i)| \; / \; |\tau(g_i)|$$

For multiple gap requirements, the weighted coverage is:

$$C(d,G_t) = [\Sigma_i w_i \, c(d,g_i)] / [\Sigma_i w_i]$$

The result lies in [0,1]. If a gap requirement becomes empty after preprocessing, the draft specifies that it should be removed before retrieval. [Project draft, p. 9]

8.4 Evidence novelty

Novelty addresses redundancy. If a candidate document is highly similar to evidence already collected, it may add little useful information. For non-empty E_t, GAR defines novelty as one minus the maximum similarity to any previously collected evidence item.

```
N(d,E_t) = 1 − max_{e ∈ E_t} s(d,e)
```

Thus, a candidate similar to at least one existing evidence item receives a lower novelty score. A candidate dissimilar to all collected evidence receives a higher novelty score. At the first iteration, E_0 is empty and the draft defines N(d,∅)=1.

```
N(d,∅) = 1
```

8.5 Final GAR score

The three normalized components are combined linearly:

```
S(d | q,E_t,G_t) = α R(d,q,G_t) + β C(d,G_t) + γ N(d,E_t)
```

The weights satisfy:

```
α ≥ 0, β ≥ 0, γ ≥ 0, α + β + γ = 1
```

Because R, C, and N are each normalized, the weighted sum also lies in [0,1]. The score is not itself a probability. It is a ranking utility score: higher values indicate that a candidate simultaneously matches the retrieval need, covers the missing information, and contributes information that is not redundant with the current evidence.

8.6 Hyperparameters

Parameter	Meaning	Constraint / role
α	Weight of relevance	α ≥ 0
β	Weight of gap coverage	β ≥ 0
γ	Weight of evidence novelty	γ ≥ 0
λ	Question-versus-gap relevance balance	0 ≤ λ ≤ 1
K_0	ANN candidate-pool size	K_0 ≥ k
k	Final retrieval depth	Fixed across systems for fairness

9. Candidate Generation and Ranking

9.1 Why approximate nearest-neighbor search is needed

Scoring every document in a large corpus with the complete GAR function can be expensive. The project therefore separates candidate generation from final GAR ranking. Approximate nearest-neighbor (ANN) search retrieves a manageable candidate pool, after which the complete GAR score is computed.

The draft emphasizes that ANN here means Approximate Nearest Neighbor, not artificial neural network. The purpose is computational efficiency.

9.2 Candidate pool

Candidates are obtained using the original question and each current gap requirement:

```
C_t = ANN_K₀(q,D) ∪ ⋃ᵢ ANN_K₀(gᵢ,D)
```

Duplicates are removed. K_0 controls the candidate pool, while k controls how many documents are finally returned.

9.3 Final ranking

```
T_t = TopK_{d ∈ C_t} S(d | q,E_t,G_t)
```

This distinction is important experimentally. If another retriever such as BM25 is used to generate the candidate pool and GAR only reranks those candidates, the resulting experiment is a GAR reranker experiment. The primary formulation in the project draft instead uses ANN candidate generation followed by GAR scoring. [Project draft, pp. 12–13]

10. Full Iterative Architecture

The complete loop separates four conceptual responsibilities: state construction, candidate generation, GAR ranking, and evidence/state update.

Step	Operation	Output
1	Receive original benchmark question q	q
2	MIGRES identifies collected evidence E_t and current gap G_t	(q,E_t,G_t)
3	Embed q, gaps, and documents	vectors
4	ANN search using q and gap requirements	candidate set C_t
5	Compute R, C, and N for each candidate	three component scores
6	Combine with α, β, γ	GAR score S
7	Return top-k candidates	T_t
8	MIGRES filters/extracts/updates evidence	E_{t+1}, G_{t+1}
9	Repeat if a gap remains	next retrieval iteration

```
(q,E_t,G_t,D) → GAR → T_t → MIGRES → (E_{t+1},G_{t+1}) → GAR → …
```

This is the defining iterative behavior: the retrieval score is state-dependent. The state changes after each evidence extraction step, so the ranking can change even when the original question remains fixed.

11. Baseline Retrieval Models

11.1 BM25

BM25 is the classical sparse probabilistic baseline. It combines term frequency, inverse document frequency, and document-length normalization.

```
S_BM25(q,d) = Σ_{t∈q} IDF(t) · [ f(t,d)(k₁+1) / ( f(t,d) + k₁(1−b+b|d|/avgdl) ) ]
```

Here f(t,d) is the frequency of term t in document d; |d| is the document length; avgdl is average document length; and k_1 and b control term-frequency saturation and length normalization.

11.2 DPR

Dense Passage Retrieval represents the query and passages using dense encoders and ranks documents by vector similarity.

$$S_{DPR}(q,d) = q^{\top}d$$

DPR is important because it represents a strong supervised dense-retrieval paradigm and provides a contrast to lexical BM25.

11.3 Contriever

Contriever is an unsupervised dense retriever based on contrastive learning.

$$S_{Contriever}(q,d) = sim(q,d)$$

It gives the comparison a dense retriever that is not trained in the same supervised manner as DPR.

11.4 ColBERTv2

ColBERTv2 uses late interaction: query and document tokens receive contextualized representations and the system computes token-level interactions.

$$S_{ColBERT}(q,d) = \Sigma_i \max_j sim(q_i,d_j)$$

The max operation for each query token finds its strongest matching document token, and the query-token scores are then aggregated. This preserves more fine-grained interaction than a single-vector document representation.

11.5 SPLADE

SPLADE creates sparse neural representations over a vocabulary and performs retrieval through sparse vector interaction.

$$S_{SPLADE}(q,d) = s_q^{\top} s_d$$

This provides a neural-sparse baseline and helps test whether GAR's benefit depends merely on moving away from lexical retrieval.

12. Experimental Protocol

12.1 Training versus validation versus test

The proposed GAR formulation has no trainable parameters in the primary implementation. The embedding model is fixed. Nevertheless, GAR contains scoring hyperparameters that must be selected without contaminating the final test set.

The proposed protocol therefore uses an official development/validation split for hyperparameter selection and a held-out test split for the final evaluation. The selected configuration is frozen before test evaluation. [Project draft, pp. 14–15]

12.2 Bayesian optimization

The draft proposes Bayesian optimization for the parameter set $\theta_{GAR} = \{\alpha,\beta,\gamma,\lambda,K_0\}$. The search space obeys the score constraints and uses a predefined discrete set for K_0.

$$\theta_{GAR} = \{\alpha,\beta,\gamma,\lambda,K_0\}$$

$$\alpha,\beta,\gamma \geq 0;\ \alpha+\beta+\gamma=1;\ 0\leq\lambda\leq1;\ K_0 \in \{50,100,200,500\}\ \text{(example search set)}$$

The exact search space and random seed should be recorded before the final test evaluation. The validation objective should be selected before inspecting test results.

12.3 Fair comparison

- Use the same question set.
- Use the same searchable corpus.
- Use the same final retrieval depth k.
- Keep MIGRES prompts, filtering, extraction, answer generation, and stopping conditions fixed wherever possible.
- Use the same evaluation scripts and record random seeds where applicable.
- Record the exact checkpoint, ANN index type, hardware, software versions, and corpus release.

13. Datasets

The project draft proposes six datasets used in the MIGRES evaluation: WikiHop, HotpotQA, MuSiQue, Natural Questions, TriviaQA, and StrategyQA. The draft emphasizes that exact counts should be taken from the downloaded release because preprocessing and release versions can change dataset sizes. [Project draft, pp. 15–17]

Dataset	Why it is relevant
WikiHop	Multi-document and multi-hop reasoning; requires combining evidence across documents.
HotpotQA	Multi-hop QA with supporting facts; FullWiki setting provides a larger retrieval problem.
MuSiQue	Compositional multi-hop questions, designed to reduce shortcut solutions.
Natural Questions	Real search questions with Wikipedia annotations; useful for open-domain retrieval.
TriviaQA	Large heterogeneous evidence collection with substantial lexical variation.
StrategyQA	Questions with implicit reasoning strategies and supporting evidence.

During test-time retrieval, gold answers and gold supporting evidence must not be used to construct G_t, E_t, or the ANN candidate pool. They are reserved for evaluation. [Project draft, pp. 16–17]

14. Evaluation Metrics — Mathematical Detail

14.1 Recall@k

Recall@k measures how much of the relevant evidence appears within the first k retrieved documents.

```
Recall@k = |Relevant evidence ∩ Top-k| / |Relevant evidence|
```

For GAR, this is particularly meaningful because the method is designed to retrieve information that is still missing.

14.2 Precision@k

```
Precision@k = |Relevant documents in Top-k| / k
```

Precision measures the fraction of retrieved documents that are relevant. Recall and precision should be interpreted together because a system can increase recall by returning many documents.

14.3 Mean Reciprocal Rank

```
MRR = (1/N) Σᵢ (1/rankᵢ)
```

rank_i is the position of the first relevant document for question i. MRR is therefore sensitive to how early the first useful document appears.

14.4 nDCG@k

```
DCG@k = Σ_{i=1}^k [(2^{rel_i} − 1) / log₂(i+1)]
```

```
nDCG@k = DCG@k / IDCG@k
```

nDCG is useful when relevance can be graded rather than treated as simply relevant or irrelevant.

14.5 Downstream answer quality

The project should use the official benchmark evaluation script whenever available. Typical measures include Exact Match, token-level F1, classification accuracy, and supporting-fact/evidence quality where annotations permit.

```
EM_i = 1 if predicted answer matches an accepted normalized answer; otherwise 0
```

```
EM = (1/N) Σᵢ EM_i
```

```
F1 = 2PR/(P+R)
```

These metrics answer a different question from retrieval metrics: a retriever may improve document recall without improving the final answer, so both levels should be reported.

14.6 Efficiency and cost

```
Cost(q) = T_in(q)·p_in + T_out(q)·p_out
```

The complete evaluation should distinguish local retrieval computation from external LLM costs. The draft proposes recording retrieval latency, ANN indexing time, index memory, number of retrieval iterations, number of documents scored by GAR, number of LLM calls, input/output tokens, and API cost. [Project draft, pp. 20–21]

15. Ablation Studies

Ablation is essential because the full score combines three components. Without ablations, an observed improvement would not establish which component produced it.

15.1 Remove relevance

```
S_{−R}(d) = β C(d,G_t) + γ N(d,E_t)
```

15.2 Remove coverage

```
S_{−C}(d) = α R(d,q,G_t) + γ N(d,E_t)
```

15.3 Remove novelty

```
S_{−N}(d) = α R(d,q,G_t) + β C(d,G_t)
```

15.4 Diagnostic variants

```
GAR_Q-only: use only question relevance
```

```
GAR_{Q+G}: use question + gap relevance, without novelty
```

The diagnostic variants are useful because they separate three possible explanations: improvement from simply using the gap, improvement from lexical coverage, and improvement from explicit redundancy avoidance.

15.5 Candidate-pool sensitivity

Because K_0 controls how many documents receive the full GAR score, it should be varied on validation data. The draft proposes K_0 values such as 50, 100, 200, and 500 and recommends reporting both retrieval quality and latency. [Project draft, p. 19]

16. Novelty and Prior-Art Boundary

The present logbook should not state that the GAR equation is definitively novel merely because it was independently formulated. The uploaded draft itself says that the individual ingredients—semantic relevance, information coverage, and novelty—are grounded in previous work. [Project draft, pp. 1–2]

The scientifically defensible claim at this stage is narrower: GAR is a proposed unified scoring formulation that combines question/gap relevance, explicit gap coverage, and evidence novelty inside an iterative MIGRES retrieval loop. Whether that exact formulation, or an equivalent formulation, has already appeared elsewhere must be checked systematically before publication.

16.1 Literature checks that should be performed before claiming novelty

- Search for combinations of missing-information retrieval and iterative RAG.
- Search for evidence-aware and state-aware retrieval.
- Search for gap-aware, coverage-aware, and novelty-aware document ranking.
- Search for MMR-style relevance–novelty ranking applied to iterative RAG.
- Search for retrieval methods conditioned on previously retrieved evidence.
- Search for methods that explicitly model unresolved information requirements.
- Check papers, arXiv, ACL Anthology, SIGIR/CIKM/WWW/NeurIPS/ICLR/EMNLP/NAACL/ACL proceedings, dissertations, and technical reports.

A particularly important control is a comparison against a strong retriever plus established diversity/novelty reranking. Otherwise, a reviewer could reasonably ask whether the observed benefit comes from a new gap-aware mechanism or from the already-known principle of avoiding redundant evidence.

17. Implementation Plan

1. Reproduce the MIGRES + BM25 pipeline and verify the retrieval interface.
2. Confirm exactly how E_t and G_t are exposed at each retrieval iteration.
3. Download and version the six datasets and the exact searchable corpora.
4. Create fixed document embeddings using the selected sentence-transformer model.
5. Build and version the ANN index.
6. Implement the cosine-similarity utilities and token-normalization pipeline.
7. Implement gap relevance, coverage, novelty, and final GAR scoring as independently testable functions.
8. Unit-test edge cases: empty evidence, empty gap requirement, duplicate candidates, zero-length token sets, and candidate-pool smaller than k.

9 Run validation experiments and select α, β, γ, λ, and K_0.

10 Freeze the selected configuration.

11 Run all baselines under the same retrieval depth and corpus.

12 Run ablations and candidate-pool sensitivity tests.

13 Evaluate retrieval, downstream QA, latency, memory, and cost.

14 Record exact software versions, hardware, random seeds, checkpoints, and configuration files.

17.1 Recommended unit tests

Test	Expected behavior
$E_t = \emptyset$	Novelty is exactly 1 for every candidate.
G_t has one valid gap	Gap relevance and coverage reduce to single-gap scores.
One gap is empty after preprocessing	That invalid requirement is removed before scoring.
Duplicate ANN results	Candidate union removes duplicates.
$\alpha+\beta+\gamma \neq 1$	Configuration should be rejected or normalized before evaluation.
$K_0 < k$	Invalid configuration; enforce $K_0 \geq k$.
Candidate identical to existing evidence	Novelty approaches 0 under the normalized similarity definition.

18. Risks, Limitations, and Controls

Risk	Why it matters	Control
Novelty claim too broad	Relevance/coverage/diversity are established ideas.	Perform systematic prior-art search and state contribution narrowly.
ANN candidate recall bottleneck	A relevant document absent from C_t cannot be recovered by GAR ranking.	Measure candidate recall and vary K_0.
Lexical coverage mismatch	Token overlap may miss paraphrased concepts.	Compare with semantic coverage variants in future work.
Hyperparameter overfitting	Validation tuning can overfit the development set.	Freeze configuration before test; report search space and seed.
MIGRES confounding	Changes outside retrieval can affect results.	Keep non-retrieval components fixed.
Dataset leakage	Gold evidence in retrieval would invalidate results.	Strictly separate annotations from test-time retrieval.
Long-context limitations	More retrieved text does not always improve generation.	Measure downstream quality and context length.
Computational overhead	GAR scores more candidates than simple retrieval.	Report latency, K_0, scored candidates, memory, and cost.

19. Expected Scientific Contribution

The intended contribution is methodological and empirical rather than architectural. Methodologically, the project defines a state-aware retrieval score for iterative information seeking. Empirically, it proposes a controlled evaluation of whether incorporating missing information and accumulated evidence changes retrieval quality and downstream answer quality.

The project should report results neutrally. It is possible that GAR will improve some datasets, show no improvement on others, or introduce a quality–latency trade-off. A negative result would still be informative if the experimental protocol is controlled and the analysis identifies when gap-aware retrieval helps or fails.

20. Future Extensions

- Learn gap weights w_i rather than using equal weights.
- Replace lexical coverage with semantic or learned evidence coverage.
- Use a stronger embedding model and compare embedding sensitivity.
- Study alternative novelty functions, including submodular or MMR-style objectives.
- Jointly optimize candidate generation and state-aware reranking.
- Investigate adaptive stopping based on residual information gap.
- Extend from document-level ranking to passage- or sentence-level retrieval.
- Evaluate on broader retrieval benchmarks such as BEIR in addition to multi-hop QA.
- Study citation correctness and evidence attribution.
- Investigate whether GAR can reduce the number of retrieved documents while preserving downstream answer quality.

21. Logbook Entries — Project Journey

The following entries are written as a narrative record rather than as claims about exact calendar dates. Insert actual dates, meeting numbers, screenshots, code commits, and supervisor comments in the institutional version.

Entry 01 — Problem discovery

The project began with an interest in Retrieval-Augmented Generation and the role of information retrieval in supplying evidence to language models. The initial problem was broad: improve retrieval for knowledge-intensive questions. The literature immediately showed that retrieval quality is not only about the generator; the retriever determines what evidence enters the context.

Entry 02 — Narrowing the problem

The problem was narrowed from generic RAG to iterative and multi-hop retrieval. In these tasks, the system does not remain in the same information state after the first retrieval. The question may stay fixed while the evidence state changes.

Entry 03 — Reading MIGRES

MIGRES provided the key framework for thinking about this state change. Its missing-information mechanism explicitly represents what information is still needed. This suggested that the retrieval algorithm could use that state rather than treating every iteration as an independent query.

Entry 04 — Identifying the retrieval gap

A conventional relevance score can repeatedly favor documents that are strongly related to the original question. The project therefore considered a second signal: whether a candidate is relevant to the current missing information.

Entry 05 — Considering evidence coverage

Semantic relevance alone does not necessarily mean that a candidate contains the missing fact. A coverage term was therefore introduced to measure how much of the gap requirement is represented in the candidate.

Entry 06 — Considering redundancy

Once evidence has been collected, another candidate may be highly similar to what is already known. A novelty term was therefore added so that the retrieval process can favor complementary evidence rather than repeated evidence.

Entry 07 — Formulating GAR

The three signals were combined into a normalized weighted score: relevance, coverage, and novelty. The formulation was kept parameter-light and non-trainable in the primary design.

Entry 08 — Designing candidate generation

Because full scoring over an entire corpus can be expensive, approximate nearest-neighbor search was selected for candidate generation. GAR then performs the final state-aware ranking over the candidate pool.

Entry 09 — Designing fair experiments

The project was structured so that the retrieval algorithm is the main experimental variable. BM25, DPR, Contriever, ColBERTv2, and SPLADE are retained as comparison systems while the surrounding MIGRES pipeline is kept fixed wherever possible.

Entry 10 — Designing ablations

Ablation studies were added to distinguish whether performance comes from ordinary relevance, information-gap modeling, coverage, or novelty. Candidate-pool sensitivity was also included because GAR cannot rank a relevant document that ANN candidate generation fails to retrieve.

Entry 11 — Reproducibility

The final stage of the design is a reproducibility record: dataset release, corpus version, checkpoints, embedding model, ANN index, hyperparameters, software versions, hardware, seeds, prompts, and evaluation scripts must all be recorded.

Entry 12 — Publication readiness

Before publication, the remaining work is empirical validation plus systematic prior-art checking. The formulation should be described as proposed until equivalent methods have been ruled out through literature and patent/technical-report searches.

22. Conclusion

The project developed from a conventional RAG retrieval question into a more specific iterative-retrieval problem. The key conceptual shift is from ranking documents only by their relevance to the original question toward ranking them according to the current information state.

GAR formalizes that shift with three signals: R(d,q,G_t) for relevance to the question and gap, C(d,G_t) for missing-information coverage, and N(d,E_t) for evidence novelty. The final score is a constrained weighted combination of those components. [Project draft, pp. 8–10]

```
S(d | q,E_t,G_t) = αR(d,q,G_t) + βC(d,G_t) + γN(d,E_t)
```

The project is therefore best understood as a hypothesis-driven retrieval study. The central empirical question is not whether the equation looks plausible, but whether state-aware retrieval measurably improves evidence acquisition and downstream answering under controlled comparisons.

For a final academic submission, the logbook should be updated with actual dates, implementation screenshots, Git/experiment identifiers, supervisor feedback, validation tables, final results, statistical tests, and the exact prior-art search performed.

References

1. Wang, K., Duan, F., Li, P., Wang, S., & Cai, X. (2025). LLMs Know What They Need: Leveraging a Missing Information Guided Framework to Empower Retrieval-Augmented Generation. Proceedings of COLING 2025, 2379–2400. https://aclanthology.org/2025.coling-main.163/
2. Yadav, V., Bethard, S., & Surdeanu, M. (2020). Unsupervised Alignment-based Iterative Evidence Retrieval for Multi-hop Question Answering. Proceedings of ACL 2020, 4514–4525. https://aclanthology.org/2020.acl-main.414/
3. Pickett, M., Hartman, J., Bhowmick, A. K., Alam, R.-u., & Vempaty, A. (2024). Better RAG Using Relevant Information Gain. arXiv:2407.12101. https://arxiv.org/abs/2407.12101
4. Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP-IJCNLP 2019, 3982–3992. https://aclanthology.org/D19-1410/
5. Johnson, J., Douze, M., & Jégou, H. (2017). Billion-scale Similarity Search with GPUs. arXiv:1702.08734. https://arxiv.org/abs/1702.08734
6. Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in Information Retrieval, 3(4), 333–389. https://doi.org/10.1561/1500000019
7. Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W.-t. (2020). Dense Passage Retrieval for Open-Domain Question Answering. EMNLP 2020, 6769–6781. https://aclanthology.org/2020.emnlp-main.550/
8. Izacard, G., Caron, M., Hosseini, L., Riedel, S., Bojanowski, P., Joulin, A., & Grave, E. (2022). Unsupervised Dense Information Retrieval with Contrastive Learning. EMNLP 2022, 2665–2679. https://aclanthology.org/2022.emnlp-main.738/
9. Santhanam, K., Khattab, O., Saad-Falcon, J., Potts, C., & Zaharia, M. (2022). ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction. NAACL-HLT 2022, 3715–3734. https://aclanthology.org/2022.naacl-main.272/
10. Formal, T., Lassance, C., Piwowarski, B., & Clinchant, S. (2021). SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking. arXiv:2107.05720. https://arxiv.org/abs/2107.05720
11. Snoek, J., Larochelle, H., & Adams, R. P. (2012). Practical Bayesian Optimization of Machine Learning Algorithms. NeurIPS 25. https://arxiv.org/abs/1206.2944
12. Welbl, J., Stenetorp, P., & Riedel, S. (2018). Constructing Datasets for Multi-hop Reading Comprehension Across Documents. TACL, 6, 287–302. https://aclanthology.org/Q18-1021/
13. Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W. W., Salakhutdinov, R., & Manning, C. D. (2018). HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. EMNLP 2018, 2369–2380. https://aclanthology.org/D18-1259/
14. Trivedi, H., Balasubramanian, N., Khot, T., & Sabharwal, A. (2022). MuSiQue: Multihop Questions via Single-hop Question Composition. TACL, 10, 539–554. https://aclanthology.org/2022.tacl-1.31/
15. Kwiatkowski, T., et al. (2019). Natural Questions: A Benchmark for Question Answering Research. TACL, 7, 452–466. https://aclanthology.org/Q19-1026/
16. Joshi, M., Choi, E., Weld, D., & Zettlemoyer, L. (2017). TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. ACL 2017, 1601–1611. https://aclanthology.org/P17-1147/
17. Geva, M., Khashabi, D., Segal, E., Khot, T., Roth, D., & Berant, J. (2021). Did Aristotle Use a Laptop? A Question Answering Benchmark with Implicit Reasoning Strategies. TACL, 9, 346–361. https://aclanthology.org/2021.tacl-1.21/
18. Järvelin, K., & Kekäläinen, J. (2002). Cumulated Gain-based Evaluation of IR Techniques. ACM Transactions on Information Systems, 20(4), 422–446. https://doi.org/10.1145/582415.582418
19. Guu, K., Lee, K., Tung, Z., Pasupat, P., & Chang, M.-W. (2020). REALM: Retrieval-Augmented Language Model Pre-Training. ICML 2020. https://arxiv.org/abs/2002.08909

20. Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020. https://arxiv.org/abs/2005.11401

21. Borgeaud, S., et al. (2022). Improving Language Models by Retrieving from Trillions of Tokens. ICML 2022. https://arxiv.org/abs/2112.04426

22. Izacard, G., et al. (2022). Atlas: Few-shot Learning with Retrieval Augmented Language Models. JMLR / arXiv:2208.03299. https://arxiv.org/abs/2208.03299

23. Xiong, L., et al. (2021). Approximate Nearest Neighbor Negative Contrastive Learning for Dense Text Retrieval. ICLR 2021. https://github.com/microsoft/ANCE

24. Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., & Wei, F. (2022). Text Embeddings by Weakly-Supervised Contrastive Pre-training. arXiv:2212.03533. https://arxiv.org/abs/2212.03533

25. Sun, F., et al. (2019). PullNet: Open Domain Question Answering with Iterative Retrieval on Knowledge Bases and Text. EMNLP-IJCNLP 2019. https://aclanthology.org/D19-1242/

26. Li, X., et al. (2021). HopRetriever: Retrieve Hops over Wikipedia to Answer Complex Questions. AAAI 2021. https://ojs.aaai.org/index.php/AAAI/article/view/17568

27. Trivedi, H., Balasubramanian, N., Khot, T., & Sabharwal, A. (2023). Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. ACL 2023. https://aclanthology.org/2023.acl-long.557/

28. Gao, L., Ma, X., Lin, J., & Callan, J. (2022). Precise Zero-Shot Dense Retrieval without Relevance Labels. arXiv:2212.10496. https://arxiv.org/abs/2212.10496

29. Wang, L., Yang, N., & Wei, F. (2023). Query2doc: Query Expansion with Large Language Models. EMNLP 2023, 9414–9423. https://aclanthology.org/2023.emnlp-main.585/

30. Jiang, Z., et al. (2023). Active Retrieval Augmented Generation. EMNLP 2023. https://aclanthology.org/2023.emnlp-main.495/

31. Asai, A., et al. (2024). Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. ICLR 2024. https://arxiv.org/abs/2310.11511

32. Yan, S.-Q., et al. (2024). Corrective Retrieval Augmented Generation. arXiv:2401.15884. https://arxiv.org/abs/2401.15884

33. Jeong, S., et al. (2024). Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. NAACL 2024. https://aclanthology.org/2024.naacl-long.389/

34. Carbonell, J., & Goldstein, J. (1998). The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries. SIGIR 1998.

35. Thakur, N., et al. (2021). BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. NeurIPS Datasets and Benchmarks 2021. https://arxiv.org/abs/2104.08663

36. Petroni, F., et al. (2021). KILT: A Benchmark for Knowledge Intensive Language Tasks. NAACL 2021. https://aclanthology.org/2021.naacl-main.200/

37. Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2023). RAGAS: Automated Evaluation of Retrieval Augmented Generation. arXiv:2309.15217. https://arxiv.org/abs/2309.15217

38. Saad-Falcon, J., et al. (2024). ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems. NAACL 2024. https://aclanthology.org/2024.naacl-long.20/

39. Niu, C., et al. (2024). RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. ACL 2024. https://aclanthology.org/2024.acl-long.585/

40. Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the Middle: How Language Models Use Long Contexts. TACL, 12, 157–173. https://doi.org/10.1162/tacl_a_00638

Appendix A. Compact Formula Sheet

1	$(q,D) \rightarrow d_1,\ldots,d_k$	Generic retrieval mapping
2	$State_t=(q,E_t,G_t,D)$	Iterative retrieval state
3	$\varphi:T\rightarrow R^p$	Text embedding
4	$s(x,y)=½[1+\cos(x,y)]$	Normalized semantic similarity

5	$s_G(d,G_t)=\Sigma w_i s(d,g_i)/\Sigma w_i$	Gap relevance
6	$R=\lambda s(d,q)+(1-\lambda)s_G$	Question + gap relevance
7	$c(d,g_i)=\lvert\tau(d)\cap\tau(g_i)\rvert/\lvert\tau(g_i)\rvert$	Single-gap coverage
8	$C=\Sigma w_i c(d,g_i)/\Sigma w_i$	Aggregate coverage
9	$N=1-\max_{e\in E_t}s(d,e)$	Evidence novelty
10	$S=\alpha R+\beta C+\gamma N$	GAR score
11	$\alpha+\beta+\gamma=1$	Score-weight constraint
12	$C_t=ANN(q,D)\cup\bigcup ANN(g_i,D)$	Candidate pool
13	$T_t=TopK_{d\in C_t}S(d)$	Final retrieval
14	$Recall@k=\lvert Rel\cap Top\text{-}k\rvert/\lvert Rel\rvert$	Retrieval recall
15	$Precision@k=\lvert Rel\cap Top\text{-}k\rvert/k$	Retrieval precision
16	$MRR=(1/N)\Sigma 1/rank_i$	First-relevant rank
17	$nDCG=DCG/IDCG$	Graded ranking quality
18	$Cost=T_{in}\,p_{in}+T_{out}\,p_{out}$	External-model cost

Appendix B. Submission Checklist

- Insert student name(s), roll number(s), department, institution, guide name, and academic year.
- Replace narrative placeholders with actual project dates and meeting records.
- Attach screenshots of the implementation, retrieval index, experiments, and result tables if required by the department.
- Add actual validation/test results after experiments are complete; do not fabricate values.
- Record exact dataset releases, corpus versions, model checkpoints, software versions, hardware, and random seeds.
- Run the prior-art search before using strong novelty language in the final paper.
- If GAR is compared with MMR or other diversity rerankers, document those experiments explicitly.
- Include statistical significance and confidence intervals where appropriate.
- Keep the final bibliography synchronized with every citation appearing in the report.